

Ricardo Rey-Sáez



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Education

- 2024–present | **Ph.D. Psychology**, Universidad Autónoma de Madrid
 - Developing and validating psychometric models for the study of individual differences in experimental psychology.
- 2020–2022 | **M.S. Quantitative Research Methods for Behavioral and Health Sciences**, Complutense University of Madrid
- 2017–2020 | **B.S. Psychology**, Complutense University of Madrid

International Research Visits

- Mar–May 2026 | **Visiting Ph.D. Researcher**, Universiteit van Amsterdam
 - Supervised by Dr. Dora Matzke (see the [Amsterdam Mathematical Psychology Lab](#))

Additional Training

- 2026 | **Measuring Individual Differences in Experimental Psychology** (2.5 hours), XV Conference of the Spanish Society for Experimental Psychology (SEPEX) and II Joint Meeting of the SEPEX and the Belgian Psychological Association (BAPS)
- 2026 | **How do I preregister my study? A hands-on workshop** (2.5 hours), XV Conference of the Spanish Society for Experimental Psychology (SEPEX) and II Joint Meeting of the SEPEX and the Belgian Psychological Association (BAPS)
- 2024 | **Bayesian Cognitive Modeling: A Practical Course**, Universiteit van Amsterdam
- 2023 | **Practical Introduction to Advances in Exploratory Factor Analysis Applied to Item Analysis**, University of the Basque Country
- 2022 | **Introduction to Exploratory Graph Analysis (EGA) with R**, Zaragoza University
- 2020 | **Principles, Statistical and Computational Tools for Reproducible Data Science**, Harvard University (HarvardX) | [Verified Certificate](#)
- 2020 | **Professional Certificate in Data Science**, Harvard University
 - *Data Science: R basics* | [Verified Certificate](#)
 - *Data Science: Visualization* | [Verified Certificate](#)
 - *Data Science: Probability* | [Verified Certificate](#)
 - *Data Science: Inference and Modeling* | [Verified Certificate](#)
 - *Data Science: Productivity Tools* | [Verified Certificate](#)

- *Data Science: Wrangling* | [Verified Certificate](#)
- *Data Science: Linear Regression* | [Verified Certificate](#)

Teaching and Educational Activities

- 2025 | **Meta-analysis, Latent Class and Latent Profile Analysis**
 - Lead instructor, Advanced Statistical Training for Researchers
 - Organized by Deusto FamilyPsych, Deusto University
- 2025 | **Introduction to Bayesian Psychometrics: Applications to Factor Analysis**
 - Seminar instructor, National interfaculty training seminar
 - Organized by ASCOFAPSI (Colombian Association of Faculties of Psychology)
- 2025 | **Beyond Maximum Likelihood: Introduction to Bayesian Factor Analysis**
 - Seminar instructor, Master's program in Quantitative Research Methods
 - Organized by Autonomous University of Madrid
- 2024 | **ChatGPT and Psychometrics: Optimizing the Design, Refinement, and Data Analysis of Measurement Instruments**
 - Seminar instructor, Master's program in Quantitative Research Methods
 - Organized by Autonomous University of Madrid
- 2023–2024 | **Data Analysis in Psychology**
 - Course instructor, Undergraduate program (Bachelor's in Psychology)
 - Organized by Nebrija University
- 2023 | **Adapting Our Teaching to the Revolution of AI and ChatGPT**
 - Workshop instructor, Faculty training
 - Organized by Universidad Autónoma de Madrid
- 2023 | **How can we Adapt Our Teaching to the Revolution of AI and ChatGPT?**
 - Workshop instructor, Faculty training
 - Organized by University of Costa Rica

Textbooks

- **Rey-Sáez, R., & Estrada, E. (2023). Análisis de datos con R en Ciencias Sociales y de la Salud.** Editorial Síntesis: <https://www.sintesis.com/libro/analisis-de-datos-con-r-en-ciencias-sociales-y-de-la-salud>

Software

- **BayesianEFA.** R package for Bayesian exploratory factor analysis. Source code: <https://github.com/RicardoReySaez/BayesianEFA>
- **uSDT.** R package for hierarchical signal detection theory models in unconscious processing research. Co-developed with F. Garre-Frutos, A. Franco-Martínez, and M. A. Vadillo. Source code: <https://github.com/RicardoReySaez/uSDT>

Awards

- 2025 | **Best Oral Communication Award** | Ibero-American Association for Research on Individual Differences
 - Presentation Title: **New Hierarchical Models to Answer Old Psychometric Questions in Experimental Research: Which Task Best Measures Inhibition?**

- Prize: **MASK-5 Measurement Instrument**
- 2024 | **Dissemination and Transfer in Methodology Award** | Spanish Association of Methodology for Behavioral Sciences
 - Project **Ecuación Triunfo**
 - Funding: €1,000
- 2024 | **Julio Olea Grant** | Spanish Association of Methodology for Behavioral Sciences
 - Funding to carry out the research project **The union of psychometrics and experimental psychology: Bayesian hierarchical models with common factors for response times**
 - Funding: €2,400

Publications

1. **Rey-Sáez, R.**, & Revuelta, J. (2026). An Efficient Rotation-Sign-Permutation Algorithm to Solve Rotational Indeterminacy in Bayesian Exploratory Factor Analysis. *Bayesian Analysis*. <https://doi.org/10.1214/26-BA1611>
2. **Rey-Sáez, R.**, Franco-Martínez, A., Revuelta, J. & Vadillo, M. A. (2026). A Unified Framework for Psychometrics in Experimental Psychology: The Standardized Generalized Hierarchical Factor Model. *Psychological Methods*. Advance online publication. <https://doi.org/10.1037/met0000846>
3. Garre-Frutos, F., Martínez-López, P., Ariza, A., **Rey-Sáez, R.**, Luque, D., Vadillo, M. A., & González, F. (2026, March 22). Explicit knowledge modulates Pavlovian attentional biases. *PsyArxiv*. https://doi.org/10.31234/osf.io/uwge3_v2
4. Franco-Martínez, A., **Rey-Sáez, R.**, Shanks, D. R., Soto, D., & Vadillo, M. A. (2026). Replicating the unconscious working memory effect: A multisite preregistered study. *Neuroscience of Consciousness*. <https://doi.org/10.1093/nc/niaf046>
5. Pacheco-Romero, A. M., Martín-García, Ó., **Rey-Sáez, R.**, Boemo, T., Blanco, I., Vázquez, C., & Sánchez-López, Á. (2024). An integrative analysis of potential mechanisms of reduced positive affect in daily life in depression: an ESM study. *Cognition and Emotion*, 38(4), 587–604. <https://doi.org/10.1080/02699931.2024.2314066>
6. Gracia, E., Martín-Fernández, M., **Rey-Sáez, R.**, & Yount, K. M. (2024). Attitudes justifying partner violence against women in Latin-American and Caribbean countries: A measurement invariance study. *Psychology of Violence*, 14(2), 127–136. <https://doi.org/10.1037/vio0000497>
7. Vega-Sanz, M., **Rey-Sáez, R.**, Berástegui, A., & Sánchez-López, Á. (2024). Difficulties in pregnancy adjustment as predictors of perinatal depression: Indirect effects through the role of brooding rumination and maternal–fetal bonding. *American Journal of Perinatology*, 41(S 01), e2870–e2877. <https://doi.org/10.1055/s-0043-1776062>
8. Franco-Martínez, A., **Rey-Sáez, R.**, & Castillejo, I. (2023). The seven wonderings of large language models as psychometric designers, refiners, and analysts. *PsyArXiv*. <https://doi.org/10.31234/osf.io/kmqy5>
9. **Rey-Sáez, R.** (2022). Réplica a: Validación cruzada sobre una misma muestra: Una práctica sin fundamento. *R.E.M.A. Revista Electrónica de Metodología Aplicada*, 24(1), 41–44. <https://doi.org/10.17811/rem.a.24.1.2022.41-44>

Conference Presentations

1. **Rey-Sáez, R.** (2026). *Experimental Psychometrics*. [Symposium Organizer] XV Conference of the Spanish Society for Experimental Psychology (SEPEX) and II Joint Meeting of the SEPEX and the Belgian Psychological Association (BAPS).

2. **Rey-Sáez, R.**, Franco-Martínez, A., Revuelta, J., & Vadillo, M. A. (2026). *Don't Aggregate-then-Correlate, Model! Unleashing the Power of Generalized Hierarchical Factor Models in Experimental Psychology*. [Oral presentation] Symposium: *Experimental Psychometrics*. XV Conference of the Spanish Society for Experimental Psychology (SEPEX) and II Joint Meeting of the SEPEX and the Belgian Psychological Association (BAPS).
3. Garre-Frutos, F., Franco-Martínez, A., Vadillo, M. A., & **Rey-Sáez, R.** (2026). *A multilevel-signal detection framework for unconscious processing research*. [Oral presentation] Symposium: *Experimental Psychometrics*. XV Conference of the Spanish Society for Experimental Psychology (SEPEX) and II Joint Meeting of the SEPEX and the Belgian Psychological Association (BAPS).
4. Garre-Frutos, F., Martínez-López, P., **Rey-Sáez, R.**, Luque, D., Vadillo, M. A., & González, F. (2026). *Explicit knowledge modulates Pavlovian attentional biases*. [Oral presentation] Symposium: *Current Debates in Research on Habits*. XV Conference of the Spanish Society for Experimental Psychology (SEPEX) and II Joint Meeting of the SEPEX and the Belgian Psychological Association (BAPS).
5. Franco-Martínez, A., **Rey-Sáez, R.**, Revuelta, J., & Vadillo, M. A. (2026). *Cuando el análisis factorial empezó a contestar preguntas de psicología experimental*. [Oral presentation] Symposium: *Nuevos desarrollos en Análisis Factorial*. XIX Congreso de Metodología de las Ciencias Sociales y de la Salud.
6. **Rey-Sáez, R.**, & Revuelta, J. (2026). *Rotaciones Oblicuas en AFE Bayesiano: superando a los métodos frecuentistas en su propio terreno*. [Oral presentation] Symposium: *Nuevos desarrollos en Análisis Factorial*. XIX Congreso de Metodología de las Ciencias Sociales y de la Salud.
7. **Rey-Sáez, R.** (2026). *Simultaneous Latent Dimensionality Assessment and Factor Loading Estimation in Sparse Bayesian Exploratory Factor Analysis*. [Oral presentation] XIX Congreso de Metodología de las Ciencias Sociales y de la Salud.
8. **Rey-Sáez, R.**, Franco-Martínez, A., Revuelta, J., & Vadillo, M. A. (2026). *A Multilevel Psychometric Framework for Experimental Tasks: The Generalized Hierarchical Factor Model* [Oral presentation] 15th International Multilevel Conference.
9. Franco-Martínez, A., **Rey-Sáez, R.** & Vadillo, M. A. (2026) *How do I start planning my multisite project? A tutorial on multilevel simulation-based power and sensitivity analysis* [Oral presentation] 15th International Multilevel Conference.
10. **Rey-Sáez, R.**, Franco-Martínez, A., Revuelta, J., & Vadillo, M. A. (2025). *New Hierarchical Models to Answer Old Psychometric Questions in Experimental Research: Which Task Best Measures Inhibition?* [Oral presentation] XIV Conference – Asociación Iberoamericana para la Investigación de las Diferencias Individuales (AIIDI).
11. **Rey-Sáez, R.**, Franco-Martínez, A., Revuelta, J., & Vadillo, M. A. (2025). *Where psychometrics meets experimental psychology: Bayesian hierarchical factor models for response times*. [Oral presentation] XI Conference – European Congress of Methodology.
12. **Rey-Sáez, R.**, Revuelta, J., & Vadillo, M. A. (2025). *From random effects to common factors: Latent dimensionality assessment in experimental psychology*. [Oral presentation] XI Conference – European Congress.
13. **Rey-Sáez, R.**, Garre-Frutos, F., Franco-Martínez, A., Castillejo, I., & Vadillo, M. A. (2025). *Addressing methodological challenges in unconscious process research: A hierarchical modeling approach*. [Poster presentation] 27 Annual Meeting of the Association for Scientific Studies of Consciousness.
14. **Rey-Sáez, R.**, Garre-Frutos, F., Franco-Martínez, A., Castillejo, I., & Vadillo, M. A. (2025). *Hierarchical models in experimental psychology: A solution for measuring unconscious psychological processes*. [Poster presentation] XIV Reunión Científica sobre Atención.
15. **Rey-Sáez, R.**, Garre-Frutos, F., & Vadillo, M. Ángel. (2024). *What psychometrics can offer to experimental psychology: The multilevel model as the legacy of factor analysis*. [Oral presentation] XVIII Congreso de Metodología de las Ciencias Sociales y de la Salud.

16. **Rey-Sáez, R.**, Lecuona, Ó., & Martín-Fernández, M. (2024). *Approximate measurement invariance with multiple groups: A comparison between Bayesian SEM and the alignment method*. [Oral presentation] XVIII Congreso de Metodología de las Ciencias Sociales y de la Salud.
17. Lecuona, Ó., **Rey-Sáez, R.**, & Martín-Fernández, Manuel. (2024). *Studying invariance with multiple groups in large datasets: A comparison of bayesian SEM and alignment method*. [Oral presentation] 13th Conference of The International Test Commission.
18. **Rey-Sáez, R.**, Franco-Martínez, A., Castillejo, I., & Vadillo, M. A. (2024). *From unconscious to unreliable? Measurement error biases evidence on studies of unconscious processing*. [Poster presentation] 27 Annual Meeting of the Association for Scientific Studies of Consciousness.
19. Franco-Martínez, A., **Rey-Sáez, R.**, Shanks, D. R., Soto, D., & Vadillo, M. A. (2024). *Advances in "replicating the unconscious working memory effect: A multisite preregistered study."* [Poster presentation] 27 Annual Meeting of the Association for Scientific Studies of Consciousness.
20. Pacheco-Romero, A. M., Martín-Gracia, Ó., **Rey-Sáez, R.**, Boemo, T., Blanco, I., & Sánchez-López, Á. (2023). *Affective reactivity to daily events predicts changes in future psychopathology and is modulated by the use of different forms of rumination*. [Oral presentation] International Convention of Psychological Science (ICPS).
21. Aivar, P., **Rey-Sáez, R.**, & Vadillo, M. A. (2023). *Hard to ignore? Irrelevant distractors cannot be completely suppressed in a contextual cueing task*. [Poster presentation] 23 Annual Meeting of the Vision Sciences Society.
22. Martín-García, O., **Rey-Sáez, R.**, Boemo, T., Pacheco-Romero, A., Blanco, I., & Sánchez-López, A. (2023). *Modulatory role of motivational self-focuses on the relation between perceived stress and momentary affect: An ESM study*. [Oral presentation] Emotions Congress 2023.
23. Pacheco-Romero, A. M., Martín-Gracia, Ó., **Rey-Sáez, R.**, Boemo, T., Blanco, I., Vázquez, C., & Sánchez-López, Á. (2022). *Intra-individual mechanisms of positive affect regulation deficits in depression: An experience sampling study*. [Oral presentation] 52nd European Association for Behavioural; Cognitive Therapies.
24. **Rey-Sáez, R.**, & Nájera, P. (2022). *Investigando metodología en el multiverso de la academia: Comunidad, recursos y estrategias por y para jóvenes*. [Symposium Organizer] XVII Congreso de Metodología de las Ciencias Sociales y de la Salud.
25. **Rey-Sáez, R.** (2022). *Investigación, paso 1: ¿Cómo y dónde busco información? Nuevas herramientas para la detección eficiente de bibliografía de interés*. [Oral presentation] XVII Congreso de Metodología de las Ciencias Sociales y de la Salud.
26. **Rey-Sáez, R.** (2022). *Approximate measurement invariance: Assessing the accuracy and sensitivity of alignment optimization under partial invariance with multiple groups*. [Oral presentation] XVII Congreso de Metodología de las Ciencias Sociales y de la Salud.
27. Martín-Fernández, M., & **Rey-Sáez, R.** (2022). *Approximate measurement invariance in surveys on attitudes toward intimate partner violence in 12 Latin American countries*. [Poster presentation] XVII Congreso de Metodología de las Ciencias Sociales y de la Salud.
28. **Rey-Sáez, R.** (2022). *Approximate measurement invariance in surveys on attitudes toward intimate partner violence in 12 Latin American countries*. [Oral presentation] Research Conference of the Faculty of Psychology, Universidad Autónoma de Madrid